ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

CCS3142 MOBILE APPLICATION DEVELOPMENT

FINAL SEMESTER EXAMINATION

TUESDAY, 3 DECEMBER 2024

09:00 - 12:00 HOURS

TIME ALLOWED : WRITING – THREE HOURS : READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

NOTE: Your answers should NOT be brief. Give detailed essay style reasoned answers with structured paragraphs and headings.

Mobile application development is the process of creating software designed to run on mobile devices like smartphones and tablets. It involves a combination of creativity, technology and strategic planning to meet the growing demand for mobile solutions across various industries.

The process typically begins with identifying user needs and defining the app's purpose. Developers choose a platform, such as Android or iOS and decide whether to build a native app , a cross-platform app or a progressive web app.

Key stages in mobile app development include UI/UX design, coding and backend integration. Tools like Android Studio, Xcode and Visual Studio Code aid developers in creating functional and visually appealing apps. APIs and third-party services enhance features such as payment gateways, GPS tracking and notifications.

Testing is a crucial step to ensure the app works smoothly across devices and networks. Once deployed, apps are maintained with updates to fix bugs and introduce new features.

Mobile apps have transformed how we work, shop and communicate, becoming an indispensable part of modern life. Their development requires a balance of innovation, user-centric design and technical expertise.

a) Analyze the role of the `pubspec.yaml` file in managing dependencies and assets within a Flutter Android application. **(10 marks)**

b) Discuss the advantages and limitations of using an Android Emulator compared to physical devices for testing mobile applications during development. **(10 marks)**

c) Provide specific guidelines and examples for implementing accessibility features, such as screen readers, adaptable text sizes and color contrast adjustments to ensure usability for individuals with disabilities. **(10 marks)**

d) Evaluate the role of plugins in extending the functionality of mobile applications. Explain how they enhance the development process compared to building features from scratch? **(10 marks)**

[Total: 40 Marks]

Explain how they can enhance the development process compared to building features from scratch

SECTION B: Attempt any THREE questions in this section

Question 2

Mobile app development varies between native and hybrid approaches. Implementing microservices in mobile apps poses challenges like ensuring seamless data synchronization and maintaining API performance. Flutter's http package simplifies networking with easy implementation of GET requests for fetching data from APIs, enhancing connectivity in mobile apps.

a) Mobile app development can be categorized based on the platforms and technologies used to build the application. Each type has distinct features, benefits and challenges. Compare and contrast Native and Hybrid mobile application development. **(6 marks)**

b) Evaluate the challenges of implementing microservices in mobile applications, particularly in terms of data synchronization and API performance. **(6 marks)**

c) In Flutter, networking is handled through the http package, which allows you to fetch data from APIs or web services. Write a Flutter dart code segment that that implements a get request. **(8 marks)**

[Total: 20 Marks]

Question 3

In Flutter development, navigation styles like tab bars, drawers, and floating action buttons adapt seamlessly across screen sizes, providing intuitive and accessible user experiences. The MVC architecture fosters modular development by separating data, UI, and logic, exemplified in task management apps. Asynchronous programming with async-await ensures responsiveness during time-intensive operations like network requests.

a) Analyze the effectiveness of different widget navigation styles (e.g. tab bars, drawers, floating button) in maintaining a seamless user experience across various screen sizes. **(6 marks)**

b) Using an example, illustrate how you would implement the MVC (Model-View- Controller) architecture in a mobile application development project. **(6 marks)**

c) Asynchronous programming in Flutter allows the application to perform time-consuming operations (such as network requests, file reading, or database queries) without blocking the main thread. This ensures that the app remains responsive and provides a smooth user experience. Illustrate how this is implemented by writing a dart code spinet. **(8 marks)**

[Total: 20 Marks]

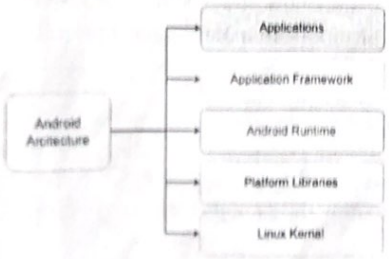
Question 4

a) In Flutter, widgets are the core building blocks of the user interface (UI). Everything you see on the screen in a Flutter app, from a simple text label to an interactive button or an entire layout, is a widget. Widgets define the structure, design and functionality of the app's UI. Write flutter dart code segments that will produce the following: **(8 marks)**



b) The hot reload feature is one of Flutter's most powerful tools, enabling developers to make changes to their code and see the results immediately without restarting the app. Identify and describe any two benefits of this feature. **(4 marks)**

c) The Android architecture shown below is a framework of components and layers that define how Android apps are developed, interact with the underlying operating system and function efficiently. Explain the functions of the components depicted in the diagram **(8 marks)**



[Total: 20 Marks]

Question 5

a) Evaluate how varying screen sizes and battery life impact the design and usability of mobile applications and propose solutions for creating a consistent user experience.

(6 marks)

b) Propose an approach to integrate obfuscation and code-splitting techniques in the mobile application development workflow to enhance security and maintainability.

(6 marks)

c) Study the code below. You are required to draw the user interface that the code will produce:

(8 marks)

```
1.  Widget build(BuildContext context) {
2.    return Scaffold(
3.      appBar: AppBar(
4.        title: Text('Flutter Widgets Example'),
5.      ),
6.      drawer: Drawer(
7.        child: ListView(
8.          padding: EdgeInsets.zero,
9.          children: <Widget>[
10.           DrawerHeader(
11.             decoration: BoxDecoration(
12.               color: Colors.blue,
13.             ),
14.             child: Text(
15.               'Drawer Header',
16.               style: TextStyle(color: Colors.white, fontSize: 24),
17.             ),
18.           ),
19.           ListTile(
20.             title: Text('Item 1'),
21.             onTap: () {
22.               Navigator.pop(context);
23.             },
24.           ),
25.           ListTile(
26.             title: Text('Item 2'),
27.             onTap: () {
28.               Navigator.pop(context);
29.             },
30.           ),
31.         ],
32.       ),
33.     ),
34.     body: Padding(
35.       padding: const EdgeInsets.all(16.0),
36.       child: Column(
```

```
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
       TextField(
        controller: _textController,
        decoration: InputDecoration(
         labelText: 'Enter Text',
         border: OutlineInputBorder(),
        ),
       ),
       SizedBox(height: 20),
       Row(
        children: [
         Checkbox(
          value: _isChecked,
          onChanged: (bool? value) {
           setState(() {
            _isChecked = value ?? false;
           });
          },
         ),
         Text('Check me'),
        ],
       ),
       SizedBox(height: 20),
       TextButton(
        onPressed: () {
         final text = _textController.text;
         final message = _isChecked
           ? "Text: $text and Checkbox is checked!"
           : "Text: $text and Checkbox is unchecked.";
         showDialog(
          context: context,
          builder: (BuildContext context) {
           return AlertDialog(
            title: Text('Your Input'),
            content: Text(message),
            actions: [
             TextButton(
              onPressed: () {
               Navigator.of(context).pop();
              },
              child: Text('Close'),
             ),
            ],
           );
          },
         );
        },
        child: Text('Submit'),
       ),
```